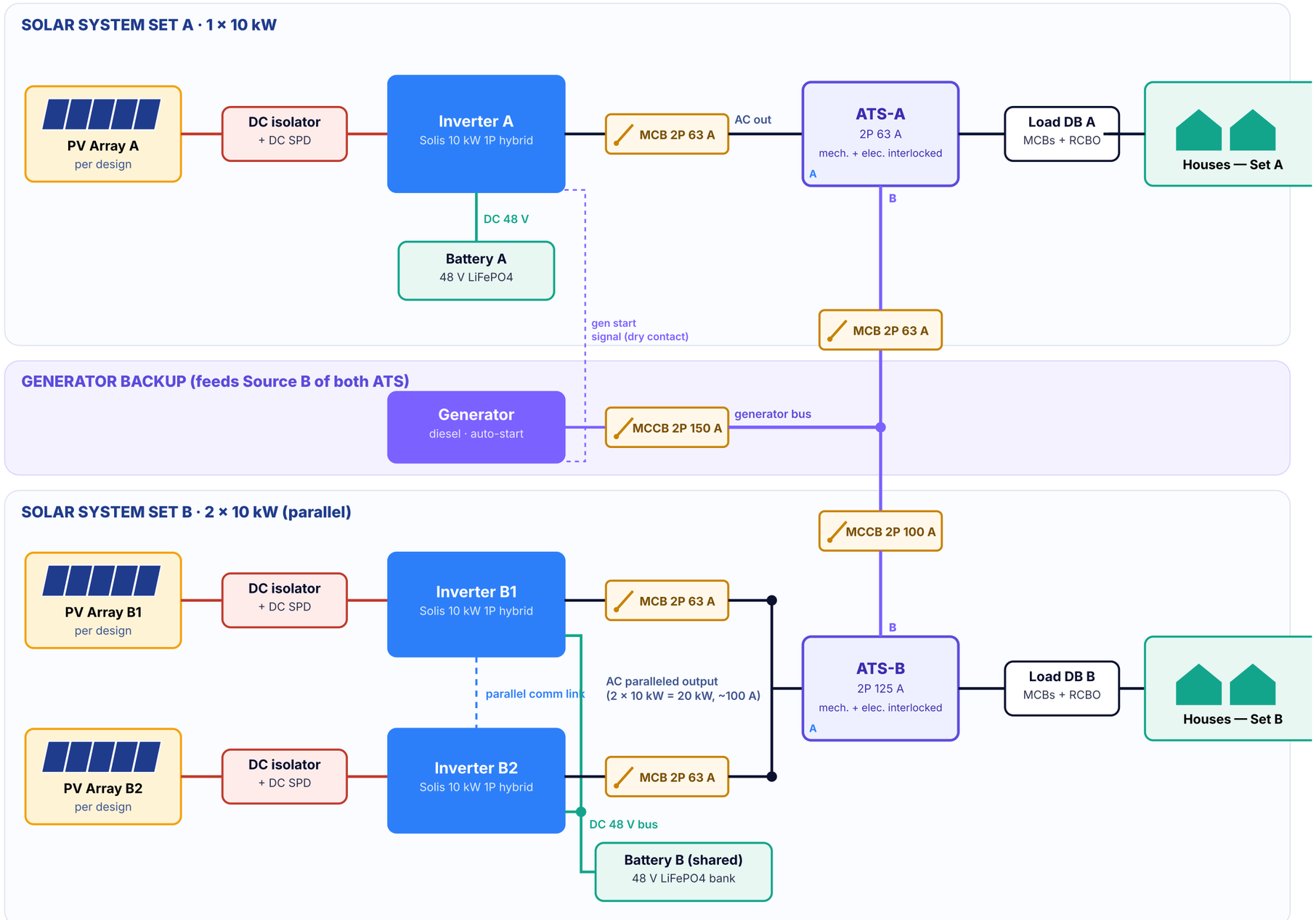
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SINGLE-LINE DIAGRAM · OFF-GRID WITH GENERATOR BACKUP

Solar Set A (1 × 10 kW) + Solar Set B (2 × 10 kW) · Generator to ATS Source B

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Legend

- DC (PV)
- DC (battery)
- AC from inverters
- AC from generator

How it operates

1. **Normal:** each solar set supplies its own houses through ATS source A.
2. **Inverter output off** (battery empty, fault or maintenance): the ATS senses loss of source A and, after a short delay, switches to source B.
3. **Generator** auto-starts on the inverter dry-contact / ATS signal and feeds Set A and Set B houses through their own ATS.
4. **Inverter back:** ATS returns to source A; generator cools down and stops.
5. **Interlock:** each ATS is mechanically and electrically interlocked, so the generator is never connected to an inverter AC output.
6. **Generator size:** ≥ 30 kVA 1-phase to carry both sets at once (or ~20 kVA with load shedding).
7. **Note:** in this layout the generator feeds loads only; it does not recharge the batteries. To charge batteries too, also wire it to the inverter GEN port.
8. **Ratings indicative** — final breaker, cable and ATS sizes per load calculation and site survey. Earth all frames, DBs and the generator to the main earth.